



X(1st Sm.)-Computer Sc.-G/(GE/CC-1)/CBCS

2022

COMPUTER SCIENCE — GENERAL

Paper : GE/CC-1

(Computer Fundamentals and Digital Logic Design)

Full Marks : 50

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question no. 1** and **any four** questions from the rest.

1. Answer **any five** questions from the following :

2×5

(a) Define BIOS.

(b) Perform the following conversions :

(i) $(10011011)_2 = (?)_{16}$

(ii) $(236)_8 = (?)_2$

(c) What is the advantage of using cache memory?

(d) Draw the truth table of a half-adder and draw its logic diagram.

(e) Write any two advantages of high level language.

(f) Find the 2's complement of the number $(101101)_2$.

(g) State any two characteristics of a multimedia software.

(h) Draw a right shift register with a suitable illustration.

2. Simplify the following functions using Karnaugh map method and implement the resultant simplified function using basic gates only.

(a) $F(A, B, C, D) = \sum(0, 2, 3, 4, 8, 12, 15)$

(b) $F(A, B, C, D) = \prod(1, 3, 4, 5, 7, 11, 12, 15)$

(3+2)+(3+2)

3. (a) Design a 3-to-8 decoder circuit.

(i) Draw the truth table.

(ii) Draw the logic diagram.

(b) Design a 3 bit binary subtractor circuit (Draw only the truth table and logic diagram). (3+3)+4

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4. (a) Design an even parity generator (Draw the truth table and logic diagram).
(b) Draw the logic diagram of a T flip-flop, and write its characteristic table. (2+2)+(3+3)
 5. (a) State the two variable De Morgan's Laws. Prove these using truth tables.
(b) What is a universal gate? Prove that NAND is a universal gate. (2+4)+(1+3)
 6. (a) Draw the logic diagram of an edge triggered RS flip-flop and draw its excitation table.
(b) Draw the logic diagram of a 4 bit ring counter. (4+2)+4
 7. Design a 4 bit asynchronous up-counter.
(a) Draw the logic diagram.
(b) Draw the truth table.
(c) Draw the timing diagram. 3+3+4
 8. Write short notes on **any two** of the following : 5×2
 - (a) System software
 - (b) Computer virus
 - (c) Compilers and Interpreters
 - (d) Seven Segment Display.
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